

SIMON

Verifiable LLM inference on hardware you don't control

Nodes sign what they computed. Anyone can check it offline.

github.com/Szpili/SIMON • MIT • Rust • libp2p

THE PROBLEM

A signature proves who wrote the number.

It never proves the number is true.

Every decentralised inference network hands you a signed claim and calls it a proof. Ours started out doing the same. The interesting engineering is in the gap between those two words.

What a receipt proves — and what it does not

Proved today

A registered node signed **this exact answer** as the result of **this job**.

- Ed25519 signature over a commitment to the output text
- Bound to the job id — a valid receipt from another run cannot be reused

Not proved

That the **declared model generated it**.

- A malicious node can sign arbitrary text with a correct digest
- Token counts and timings are signed declarations, not independent measurements

Binding model + prompt + execution + output needs an execution audit. It is not implemented, and the repository says so.

LIVE DEMO

Try to forge the receipt yourself

Tamper with the answer

Change one byte after signing.

REJECTED — signature breaks

Replay another job

A genuinely signed receipt, from a different job.

REJECTED — not proof of this work

Swap the text

Real signature, substituted answer.

REJECTED — digest covers the text

No simulated mode: with no node reachable, the demo says so instead of showing a canned result.

MEASURED, NOT PROJECTED

Verification is cheap enough to pay for

An audit is one forward pass. The job must also decode — and decoding is ~55× slower per token.

Prompt	Output	Job	Audit	Audit as % of job
500	50	7.9 s	0.63 s	8.0%
2016	200	18.8 s	1.86 s	9.9%
2011	500	87.1 s	2.07 s	2.4%
6037	500	69.1 s	5.12 s	7.4%

Lower bound: this is the forward pass alone, without activation extraction and hashing. Raw data is in the repository.

Heterogeneous on purpose — and it shows

Two GPUs, two serving stacks, two models, one network. The engine and model on each result come from the **signed receipt**, not from a label we wrote beside it.

RTX 3090 · vLLM 0.27.1

Qwen3.8-27B · full context admission gate

64.94 tok/s

A node that cannot verify its own context budget declares that, instead of pretending to check.

RTX 3080 Ti · Ollama 0.34.0

Bielik 11B · admission gate disabled, and the node says so

6.50 tok/s

Does an honest AMD node compute the same thing as an honest NVIDIA one?

If the answer is no, naive fingerprint comparison **slashes honest participants** — and every cross-vendor network has this problem, noticed or not.

Already done

CUDA baseline recorded. Notebook ships with it embedded, so it returns a verdict inside one session.

Needs MI300X

The ROCm arm under vLLM. Our backend speaks an OpenAI-compatible API — configuration, not a rewrite.

We will publish it **even if the result is inconvenient for us.**

Three of our own claims, withdrawn in public

The receipt did not bind the answer

The digest covered metadata; the text travelled unsigned.

Found while checking whether we could count tokens at all

A constant pretending to be a measurement

Throughput was **tokens / 65.0** — one card's figure, applied to every node.

Survived the code, a comment claiming "hard measurement", and a demo showing one number on two GPUs

A check the whitepaper promised and we never wrote

Both papers said the client checks a receipt is single-use. It did not.

Corrected in both languages; the ledger now gives it somewhere to be true

Status, without rounding up

Capability	State
Output binding	implemented, tested
Offline receipt verification by a third party	implemented, tested
Chunking, map-reduce, agent-to-agent chaining	working
Token counts recomputed independently	partial — backends expose no token ids
Execution correctness (TopLoc-style audit)	not implemented
Automatic slashing	deliberately off — no defensible threshold
Wash-compute resistance	unsolved

129 tests green · CI on every push, including a guard against constants posing as measurements

Idle capacity, turned into burst capacity — with a receipt you can check

Not "earn while you sleep": at 350 W a night of full load is worth less in commodity tokens than the electricity. The value is work that must not leave your machines, results that can be checked, and five GPUs at once on the morning you need them.

Code, MIT

Also in the repository

github.com/Szpili/SIMON

Whitepaper (EN and PL) · security log · ROCm notebook